

# SRI HARISH SINGARAVELAN

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[LinkedIn](#) | [GrabCAD Models](#)

## SUMMARY

Mechanical Engineer with over 3 years of experience in design, automation, and product development across the U.S. and India. Skilled in SolidWorks (CSWP), ANSYS, MATLAB, and PLM tools, with expertise in CAD modeling, FEA, GD&T, and DFM. Known for driving system efficiency improvements and accelerating prototyping cycles. Excels in cross-functional collaboration.

## EDUCATION

### Master of Science in Mechanical Engineering (Mechatronics) (3.4)

Sep 2022 – Dec 2024

Northeastern University, Boston, USA

### Bachelor of Engineering in Mechanical Engineering (3.5)

Aug 2018 – Jun 2022

LICET, Chennai, India

## PROFESSIONAL EXPERIENCE

### MEP Mechanical Designer, Ole Mexican Foods, Norcross, Georgia

Aug 2025 – Present

- Designed and simulated robotic automation workflows using RoboDK, integrating conveyors, pick-and-place robots, and vision systems to evaluate throughput and optimize production layouts.
- Led facility layout modifications and space reconfigurations for manufacturing and office areas, including wall demolition, room conversions, and operational workflow improvements.
- Developed and coordinated HVAC, plumbing, ventilation, and fire protection layouts in AutoCAD for production floors, offices, IT rooms, and distribution centers.
- Supported HVAC airflow distribution and duct sizing for shared AHU systems, accounting for occupant loads, equipment heat gain, and process requirements.
- Produced construction-ready AutoCAD drawings, XRef-based layouts, and clear scopes of work to support renovations, workstation relocations, and infrastructure upgrades.
- Identified automation opportunities in existing processes and collaborated with vendors and cross-functional teams to support integration and troubleshooting of electromechanical systems on the production floor.
- Collaborated with contractors, maintenance, and plant engineering teams through site walkthroughs, design reviews, and implementation to execute facility improvement projects efficiently.

### Mechatronics Intern, Berkshire Grey, Bedford, Massachusetts

Jan 2024 – Jun 2024

- Designed and deployed robotic end-of-arm tooling (EOAT) and mechanical subsystems for automated pick-and-place systems, improving system efficiency by ~15% through iterative design and validation.
- Engineered and validated vacuum valve prototypes (butterfly, poppet, rotary), performing rapid prototyping (3D printing), functional testing, and design optimization for reliability and cycle time improvements.
- Redesigned and released a production-ready barcode scanner mounting bracket for ABB IRB 1100 robots, improving SKU detection accuracy by ~80% and streamlining system integration.
- Developed a custom servo-driven lifecycle test rig integrating sensors, actuators, and control logic (ePLC), enabling repeatable validation of component performance across multiple test cycles.
- Built a Python-based automation script to capture real-time test data, generate performance insights, and send automated alerts, reduce manual monitoring effort and improve data accuracy.
- Created engineering drawings, BOMs, and documentation to support manufacturing, assembly, and deployment of robotic systems.

### Mechanical Engineer Intern, Symbiotic Tech Pvt Ltd, Chennai, India

Apr 2022 – Jun 2022

- Managed design documentation, BOM creation, and ECO processing through Omnify PLM, maintaining 100% compliance with internal revision control standards.
- Contributed to the design of next-generation wearable prosthetic devices, focusing on ergonomic optimization and mechanical functionality.
- Collaborated with medical staff and electrical engineers to identify optimal electrode placement, enhancing sensor position for better data quality and overall performance by 30%.
- Reduced vibration by 10% and enhanced load distribution by integrating vibration-absorbing padding and refining motor mount tolerances.
- Supported rapid prototyping using 3D printing, enabling quicker user feedback and design iterations, reducing development time by 30%.

### **Mechanical Engineer, Techcey, India**

**Jan 2021 – Dec 2021**

- Modeled and engineered plastic and sheet metal components for consumer electronics products, optimized for mass manufacturing and achieved a 12% reduction in tooling costs.
- Conducted design-for-manufacturing (DFM) reviews with suppliers and vendors, reducing part rejection rates during pilot runs by 20%.
- Applied thermal and structural analysis using SolidWorks Simulation to optimize internal component housing, resulting in a 22% improvement in heat dissipation efficiency.
- Created comprehensive documentation including BOMs, engineering drawings, and assembly instructions, ensuring smooth communication with manufacturing partners.
- Actively participated in supplier negotiations and prototype builds, accelerating the development timeline by 4 weeks.
- Led design review sessions with QA to identify failure patterns, implemented corrective measures, and enhanced product reliability by 35%.

### **Mechanical Design Intern, TVS Training and Services LTD, Chennai, India**

**Sep 2020 – Oct 2020**

- Gained hands-on experience in automotive manufacturing processes including CNC machining, welding, and assembly line operations.
- Conducted quality checks and measurements using standard metrology tools, assisting in reducing inspection time by 10%.
- Participated in lean manufacturing workshops, learning the application of Kaizen and 5S in real-world production environments.
- Assisted engineers with root cause analysis of minor production issues, improving part consistency in a simulated environment.

## **PROJECTS**

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### **LIDAR based Indoor Spatial mapping and Reconstruction, Northeastern University**

**Oct 2023 – Dec 2023**

- Designed and fabricated a LIDAR-enabled RC car for precise indoor mapping, integrating sensor data with camera feeds for enhanced depth perception.
- Applied SLAM algorithms and data fusion techniques to reconstruct accurate 3D indoor environments using LIDAR and visual data, supporting improved autonomous navigation and robotics simulation.

### **Arduino Based Obstacle Detection System, LICET**

**Jan 2022 – May 2022**

- Designed and prototyped an Arduino based obstacle detection system integrating ultrasonic sensors and embedded controls for real time obstacle detection in robotic applications.
- Implemented embedded control algorithms in C++ using Arduino IDE, integrating ultrasonic sensors and actuators, enabling real time sensor processing and actuator response.

### **Design and Development of SAE Drone Racing Quadcopter, LICET**

**Jan 2022 – May 2022**

- Engineered and spearheaded the creation of a new SAE Drone Racing team at LICET and led the design and modeling of a custom quadcopter frame from scratch in SolidWorks for intercollegiate competition.
- Presented FEA simulation findings to senior engineers, informing design modifications that enhanced the drone's responsiveness by 20% through optimized weight distribution.

## **CERTIFICATIONS**

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- [Certified SOLIDWORKS Professional - Dassault Systèmes](#)
- [Arduino: Foundations](#)
- [SOLIDWORKS: Simulation Xpress](#)
- [MATLAB - MathWorks](#)
- [Ansys Workbench - Udemy](#)
- [Python - Udemy](#)

## **SKILLS**

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- **CAD/CAE Tools:** SolidWorks (CSWP), AutoCAD, CATIA, Inventor, Revit, Creo, Fusion 360.
- **Simulation & Analysis:** ANSYS Workbench, SolidWorks Simulation, Abaqus, FEA, thermal and fluid simulations.
- **Mechanical Design:** GD&T, tolerance analysis, mechanical system design, DFM/DFA, design validation.
- **Additive Manufacturing:** 3D printing (FDM/SLA/SLS), Rapid Prototyping.
- **Control Systems & Automation:** PLCs, sensors, actuators, robotics and mechatronics integration.
- **Programming & Scripting:** MATLAB, Python, C++, ROS 2.
- **Emerging Technologies & Systems Engineering:** IoT and smart systems (basic knowledge), Digital twins and simulation modeling, Sustainable materials and energy-efficient design, Multidisciplinary system integration, model-based design.
- **Tools & Project Management:** Product Lifecycle Management (PLM): Omnify, Teamcenter SolidWorks PDM.
- **Documentation:** Technical drawings, engineering change orders (ECO), BOM creation, design reports.
- **Software:** MS Excel (advanced), Word, PowerPoint, Jira, Git, MS Project.